

Tanay Vardhan

SKILLS

Technology (Experienced):

Javascript, HTML/CSS, Angular, PHP, jQuery, MySQL, C++, Frontend development

Technology (Exposure):

Python, Node.js, Flask, SASS, React, Ionic, Vue.js, Selenium, C#, MongoDB, D3.js

Design:

Interaction design, Design thinking, User research, Photoshop, Adobe XD, InVision, Figma, Blender

Methodology and Engineering

Scrum (Agile), Kanban, Git, SVN, Star-CCM+ (CFD), SolidWorks (CAD), carbon-fiber layups

EDUCATION

University of Illinois at Urbana-Champaign

B. S. Aerospace Engineering May 2019

Minor in Computer Science

Selected Coursework:

- CS 225: Data Structures
- CS 491: Interdisciplinary Design Studio
- CS 465: Principles of User Interface Design
- CS 173: Discrete Structures
- ART 310: Design Thinking

STUDENT TEAMS & LEADERSHIP

Illini Motorsports (Formula Student Team)

Project Lead • <http://motorsports.illinois.edu>

Led project to streamline fluid simulation workflow on Linux cluster by developing Java interfaces to

- Qualitatively analyze virtual wind tunnel images across design iterations
- Configure runs for cluster by simplifying process of defining CAD geometry and setting up race environment

Illinois Autonomous Underwater Vehicle

Team Member • <http://illinoisauv.com>

Developing in-house simulation software using **Unreal Engine (C++)** to test submarine's computer vision and autonomous motion in a virtual model of competition water tank

University of Illinois Indoor Football World Cup

Captain, Team India

Led team to quarter-finals after second place in group stages with a personal tally of 3 goals and 1 assist in 5 matches

EXPERIENCE

Amadeus IT Group

Waltham, MA, USA

Software Engineering Intern • Summer 2017

- Built web application to find flights for vacations based on minimal search parameters and display results on a map-based user interface
- Developed parallel flight search voice experience for the Amazon Alexa platform to target IoT-enabled home devices
- **Technology Stack:** Angular 4, Java (Spring MVC), Node.js, AWS Lambda, HTML5/ CSS, Figma (Prototyping)

University of Illinois at Urbana-Champaign

Champaign, IL, USA

Full-Stack Developer, CEAPS • Spring 2017 - Present

- Designed and developed Digital Asia (<http://digitalasia.illinois.edu>), a repository of educational films and documentaries on East Asia
- Improved rendering performance by an average of 15 ms per page
- **Technology Stack:** Angular 4, Bootstrap 4, HTML5, CSS

Amadeus IT Group

Waltham, MA, USA

Software Engineering Intern • Summer 2016

- Developed web platform to assist implementation engineers update project statuses and improve high-level visibility for project stakeholders
- **Technology Stack:** PHP, MySQL, jQuery, HTML5, CSS, D3.js

RESEARCH & PROJECTS

Opinion Spectrum generation from vast Twitter data

- Human-Computer Interaction research project under Prof. Wai-Tat Fu, aimed at analyzing contrasting opinions on a given event or topic
- Trained open-source **TensorFlow** models to perform summarization and sentiment analysis to classify news articles on opinion and reliability
- Built web application using **Python (Flask)** to extract data using Twitter API, perform analysis using serialized "pickle" objects for classification models and visualize results.

Packeroo

Grand Prize, AngelHack Global Hackathon Series: Boston

- Skill for the Amazon Alexa to parse travel itineraries and provide useful information to assist with packing
- **Contribution:** Designed voice experience and developed companion mobile application using Ionic
- Also won the "Best Alexa Voice User Experience" prize by Amazon

Lapras

Best Use of Capital One's Nessie API, HackPrinceton Spring 2016

- Web bot to analyze transaction data and make intelligent predictions on future spending with natural language processing capabilities
- **Contribution:** Developed backend and bot logic using PHP, MySQL

Embark

Best Use of Amadeus' Innovation Sandbox, HackIllinois 2016

- Flask app to generate extensive travel itineraries with minimum input
- **Contribution:** Created high to low fidelity prototypes in Photoshop, Illustrator, developed frontend using HTML, CSS, jQuery

In-flight Entertainment Redesign

<http://tanay.io/ifeRedesign.html>

- Analysis of the shortcomings of current in-flight entertainment systems with a redesign that empathizes with users on long flights
- **Tools:** Adobe Experience Design, Google Forms (User research)